

ATIVIDADE 10

• EXEMPLO Pág. 59.

① $\sigma = 13 \text{ ml}$; $n = 30$; $\bar{x} = 210 \text{ ml}$

a) $(1-\alpha) = 95\%$, $\alpha = 5\%$; $\varphi = n-1 = 30-1 = 29$

$$IC(1-\alpha)\% = \bar{x} \pm Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

$$IC(1-\alpha)\% = 210 \pm Z_{0,05} \left(\frac{13}{\sqrt{30}} \right)$$

$$IC(95\%) = 210 \pm Z_{0,025} \cdot (2,373)$$

$$IC(95\%) = 210 \pm 1,96 \cdot (2,373)$$

$$IC(95\%) = 210 \pm 4,651$$

$$IC(95\%) = [205,349; 214,651] //$$

b) $IC(99\%) = 210 \pm Z_{0,01} \left(\frac{13}{\sqrt{30}} \right)$

$$IC(99\%) = 210 \pm Z_{0,005} (2,373)$$

$$IC(99\%) = 210 \pm 2,57 \cdot 2,373$$

$$IC(99\%) = 210 \pm 6,099$$

$$IC(99\%) = [203,901; 216,099] //$$

c) $n = \left(\frac{Z \cdot \sigma}{E} \right)^2 \rightarrow E = Z \cdot \frac{\sigma}{\sqrt{n}}$

erro máximo desejado

$$n = \left(\frac{Z_{0,025} \cdot 13}{3} \right)^2$$

$$72,131 - 30 = 42,131 \text{ copias a mais} //$$

$$n = \left(\frac{1,96 \cdot 13}{3} \right)^2 = \left(\frac{25,48}{3} \right)^2$$

$$n = 8,493^2 = 72,131$$

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